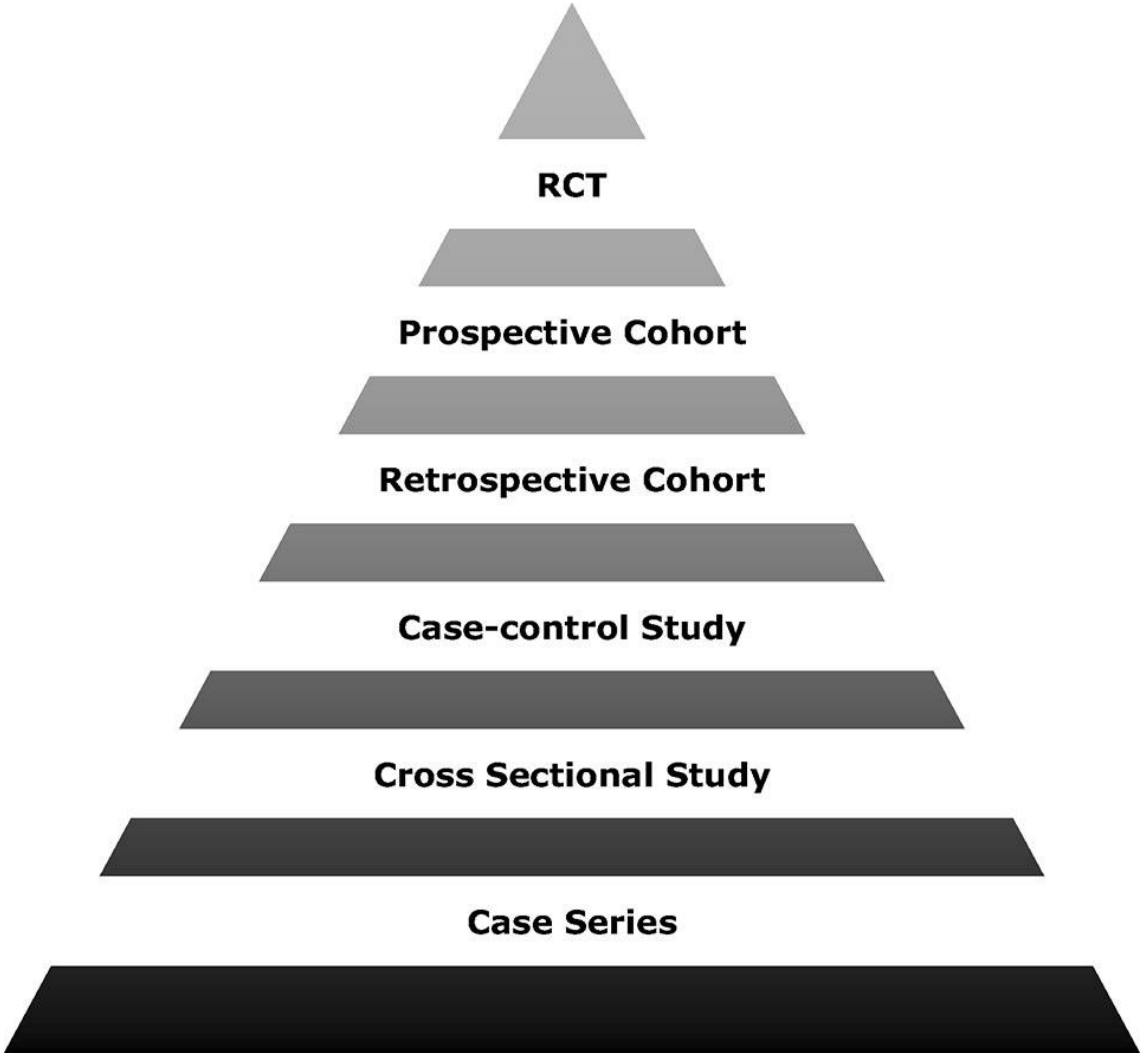
Experimental Study Design: Randomized Controlled Trial (RCT)

Hierarchy of Study Designs: Which design is best for cause and effect?



Clinical Development of Therapeutics

Preclinical Phase (Development/Validation of the therapeutic target/Pre-clinical Pharmacology and Toxicology)

Clinical Phase I (Open-label study in healthy volunteers-Biodistribution/safety and tolerability)

Clinical Phase II (Open-label/blinded randomized trial in target population to explore drug efficacy)

Clinical Phase III (properly powered Open-label/blinded randomized trial(s) in target population to estimate new treatment efficacy vs. a control)

Clinical Phase IV (Post-Marketing): Safety Surveillance/Pharmacovigilance, Effectiveness in other patient populations, RCTs for new indications

Randomized Controlled Trial (RCT)

- A prospective study comparing the effect of intervention(s) against a control group in humans
  - One or more interventions: (prophylactic, diagnostic, devices, procedures, therapeutic agents)
  - Control group: “Placebo” or “Active Drug”
    - Placebo-Controlled or Active Drug-Controlled Trials
- Gold standard for determining cause and effect relationship for beneficial drug effects

Fundamentals of an RCT

- Design of the study
- Execution of the study
- Analysis and Reporting of Results

RCT Study Design

- What are the questions that will be answered by the study?
  - **Primary Question**

    -What is the efficacy of Fidaxomicin vs. Vancomycin in curing *C. Dif* infection?
  - **Secondary Questions:**

    -For patients who were cured by treatment, how long does the cure last for and what is the risk of recurrence of infection?

    -Which drug is more tolerable?

RCT Study Design

- What are the characteristics of the study participants?

  - Inclusion (Eligibility) Criteria

  C. Dif infections defined by more than 3 unformed bowel movements in the 24 hrs prior to allocation of a treatment and either toxin A or B within 48 hrs

  - Exclusion Criteria

  Patients who had 2 or more episodes in the past 3 months

RCT Study Design

- What are some **issues** with inclusion/exclusion criteria?
  - Inclusion/exclusion criteria are set to identify subjects who will **most likely benefit** from the treatment under investigation
  - Internal vs. External Validity
    - External Validity/Generalizability:
      - How can we generalize the results from a RCT to a wide patient population who do not exactly fit the profile of patients in the RCT?
    - Internal Validity:
      - What is the role of bias (confounding, observer, selection, etc…) in the results that are reported?

RCT Study Design

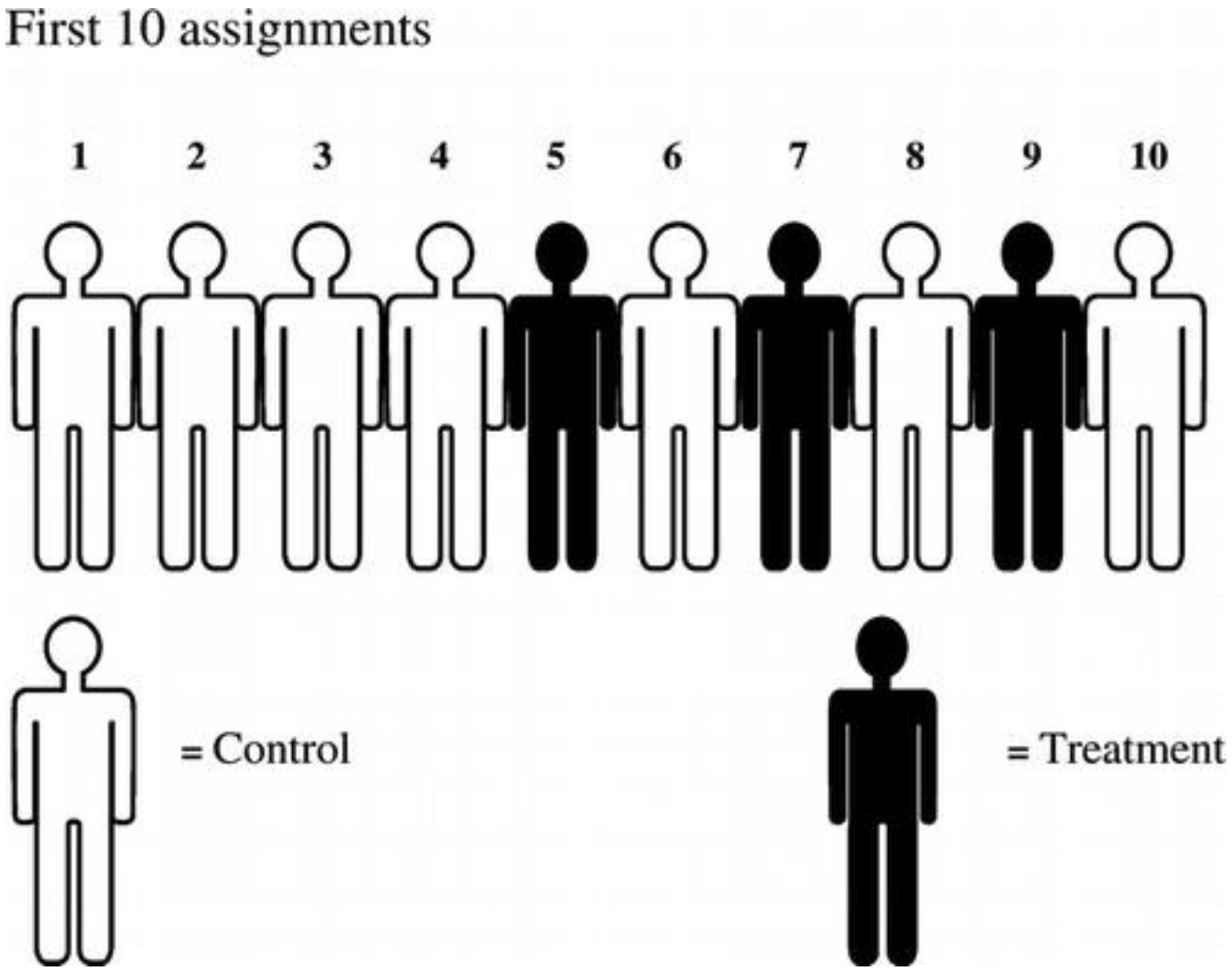
- How do RCTs handle bias?

A) Randomization (Random Assignments of Eligible Patients to either the experimental treatment or control)

-Randomization Techniques:

- Simple Randomization
- Block Randomization
- Stratified Randomization

RCT Study Design: Simple Randomization

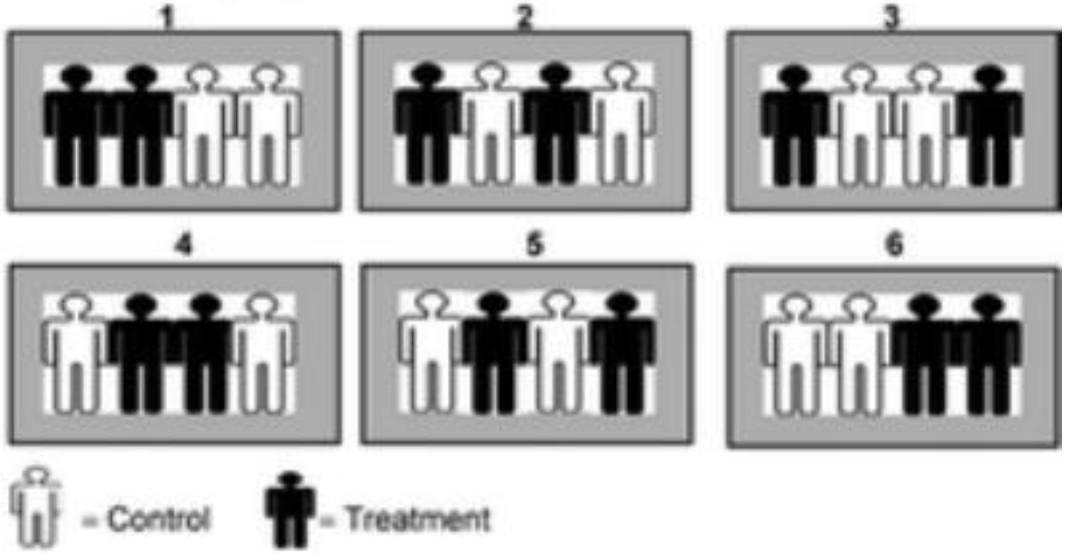


RCT Study Design: Block Randomization

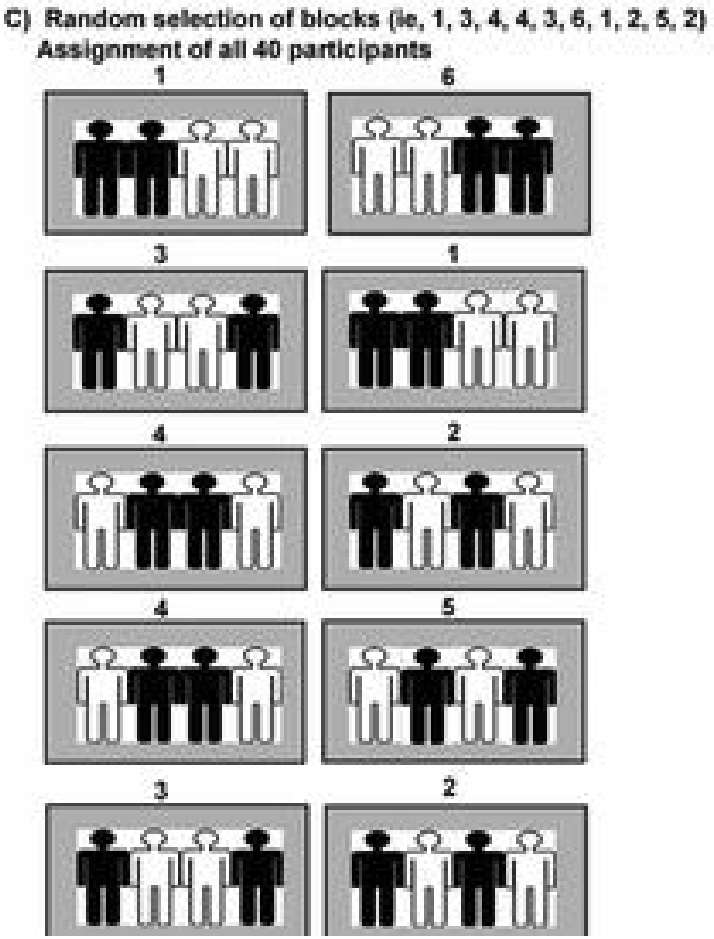
A) Block size



3) Possible balanced combinations (ie, 2 to control group, 2 to treatment group)



RCT Study Design: Block Randomization



RCT Study Design: Stratified Randomization

- What if I know that a certain characteristic may influence the response to treatment?
- e.g. previous C. Dif infection in the past 3 month may influence response to Fidaxomicin or Vancomycin and I would like to identify this influence!

- Create two strata (one for patients who had no C.Dif infection in the past three month and one for patients who had on C. Dif infection)

- Randomly assign participants in each strata to either Fidaxomicin or Vancomycin

RCT Study Design: Blinding/Masking

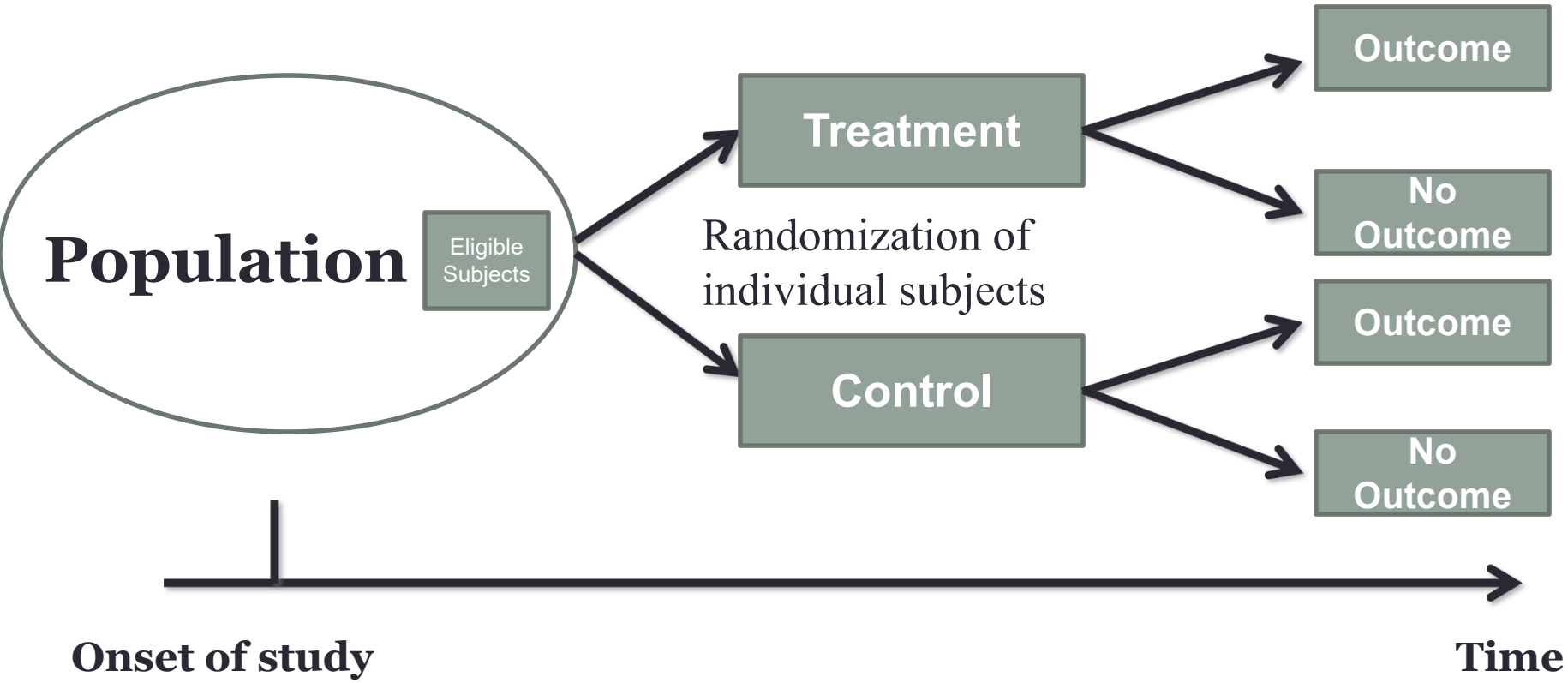
Blinding	Knowledge of Treatment Assignment	
	Patient	Investigator
None (open-label)	Yes	Yes
Single-blinded	No	Yes
Double-blinded	No	No

- An **open-label trial**, or **open trial**, is a type of **clinical trial** in which both the researchers and participants know which treatment is being administered. ... **Open-label trials** may also be uncontrolled (i.e. without a placebo group), with all participants receiving the same treatment.

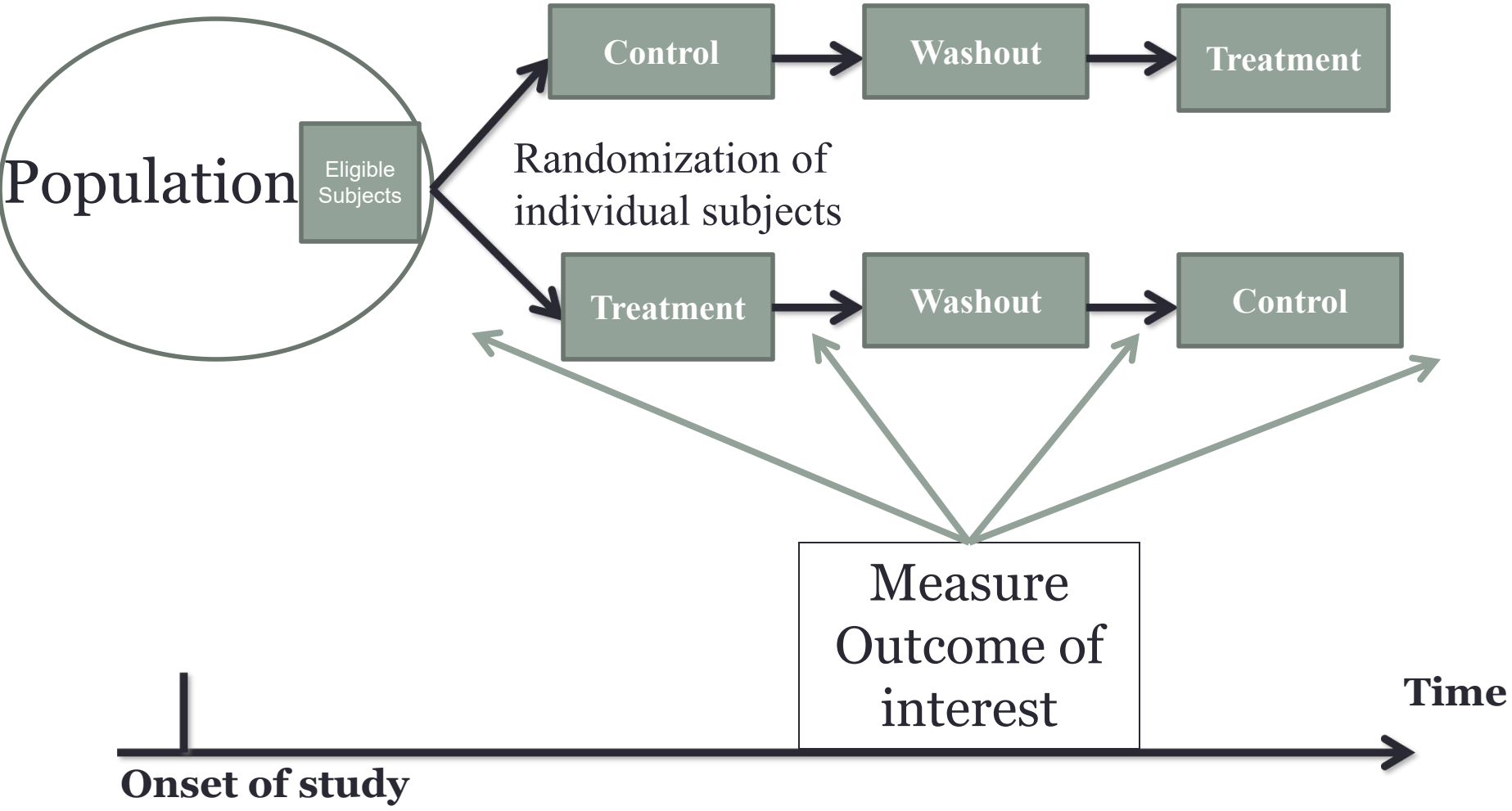
What are the benefits of blinding?

Individuals blinded	Potential benefits
Participants	Less likely to have biased psychological or physical responses to intervention
	More likely to comply with trial regimens
	Less likely to seek additional adjunct interventions
	Less likely to leave trial without providing outcome data, leading to lost to follow-up
Trial	Less likely to transfer their inclinations or attitudes to participants
Investigators	Less likely to differentially administer co-interventions
	Less likely to differentially adjust dose
	Less likely to differentially withdraw participants
	Less likely to differentially encourage or discourage participants to continue trial
Assessors	Less likely to have biases affect their outcome assessments, especially with subjective outcomes of interest

Randomized Concurrent/Parallel Trial



Randomized Crossover Trial



RCT Study Design: Group Size and Power

- Treatment Allocation Ratio
  - Treatment: Control (1:1) is the most common
  - Treatment: Control (2:1) or (3:1) can also be employed

- Group Size Determination and Power
  - Power should be set *a priori* (Usually at 80% but may go higher to 90%)

  - Based on the set power level and the minimum difference that we would like to detect (effect size), group size can be calculated.

Power calculation

Comparing Two Means: n is the required sample size per arm

$$n = \frac{2 s_p^2 \left(t_{n+m-2,1-\beta} + t_{n+m-2,1-\alpha/2} \right)^2}{\left(\mu_1 - \mu_2 \right)^2}$$

$$\approx \frac{2 s_p^2 \left(z_{1-\beta} + z_{1-\alpha/2} \right)^2}{\left(\mu_1 - \mu_2 \right)^2}$$

- For many statistical comparisons (i.e., comparing means; comparing rates; comparing censored survival times), there is an equation which involves **α**, power, **sample size**, and the **effect size**
  - You have to specify three of these quantities.
  - You then use the formula to solve for the fourth.

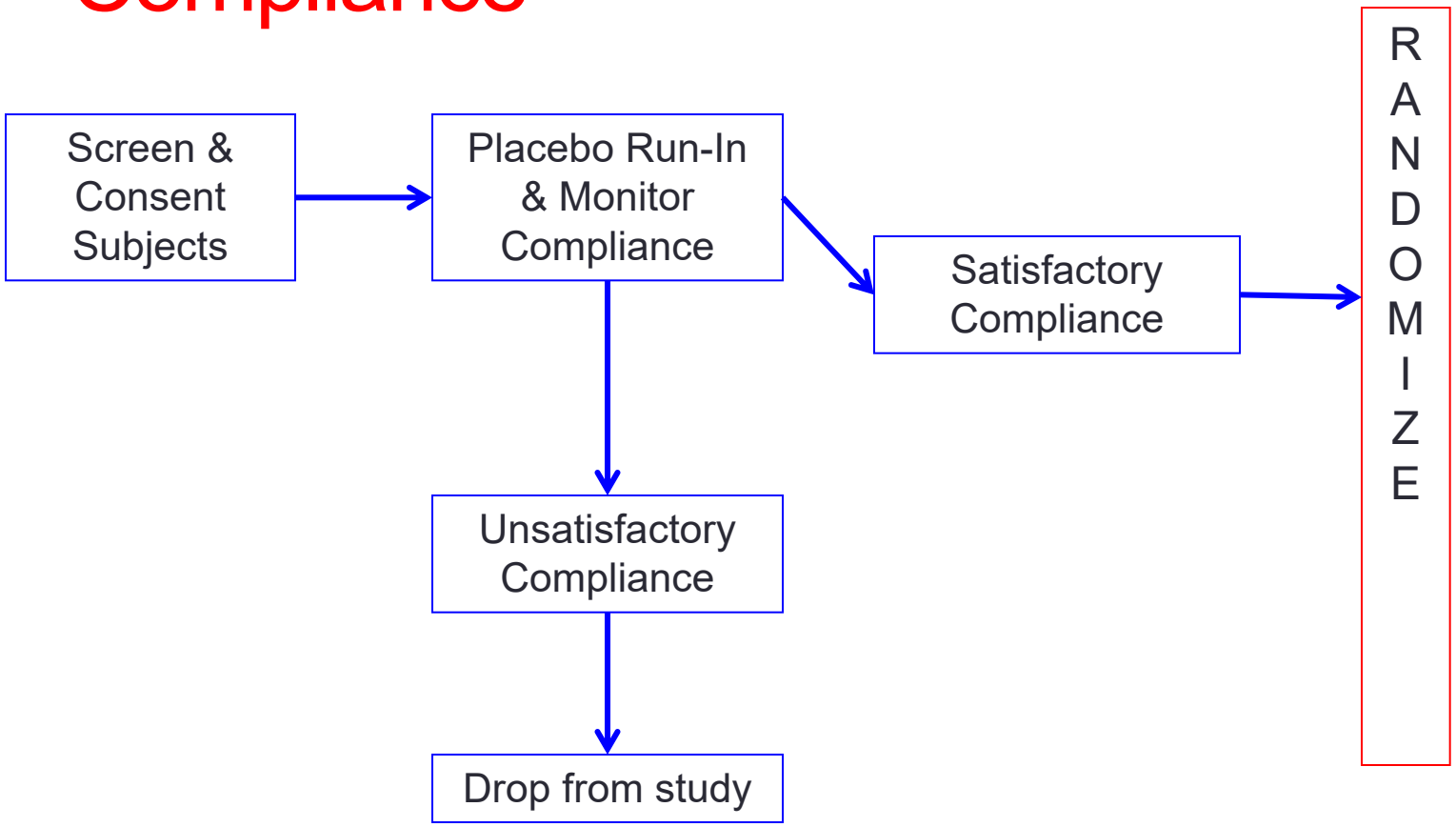
Other RCT Study Design Challenges

- Loss to follow-up
- Non-compliance
- Placebo Effect

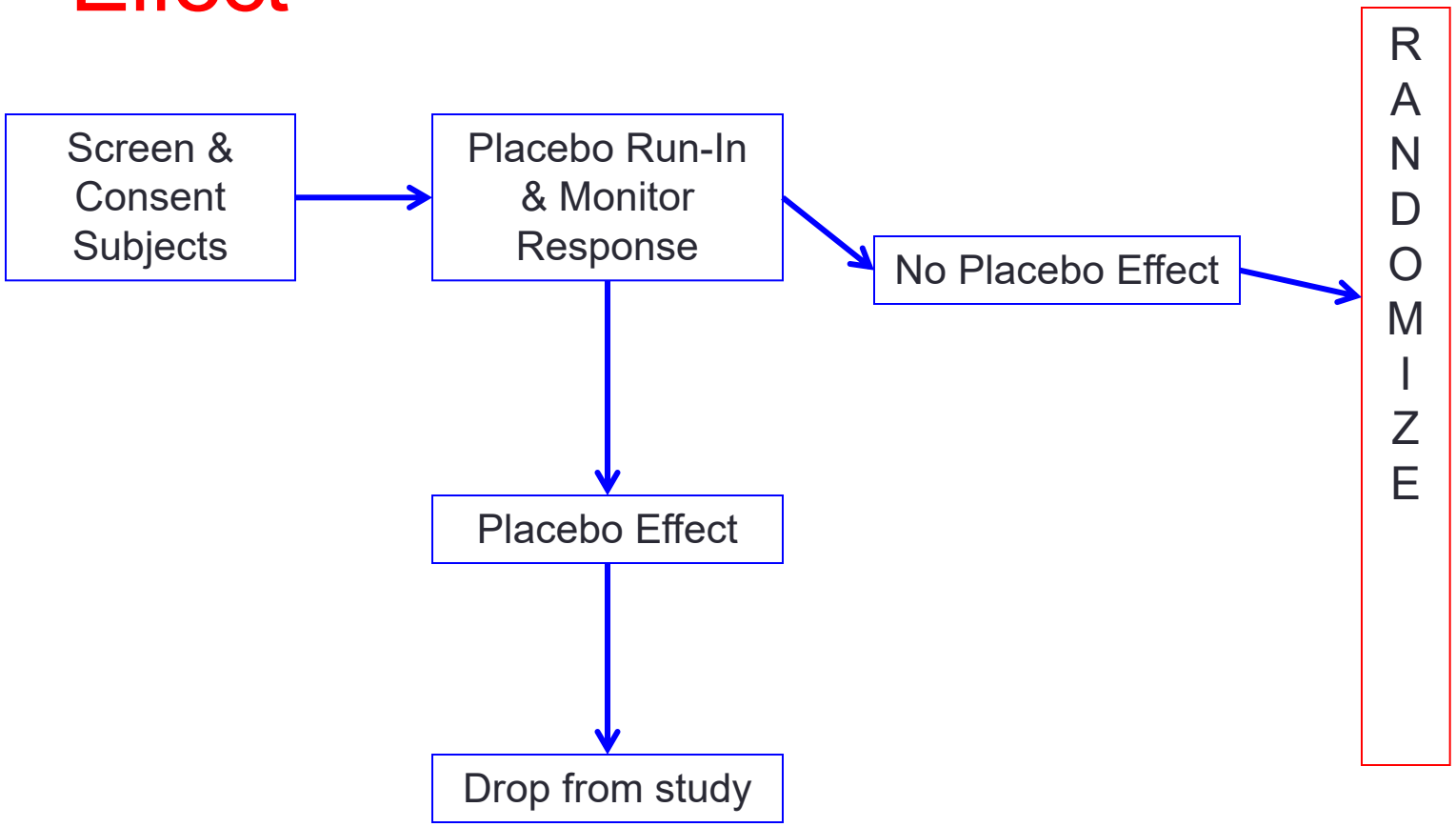
Placebo Run-In Design

Assign all eligible patients a placebo to be taken for a “brief” period of time. Patients who are “judged” compliant are enrolled into the study. This is often referred to as the “Placebo Run-In” period.

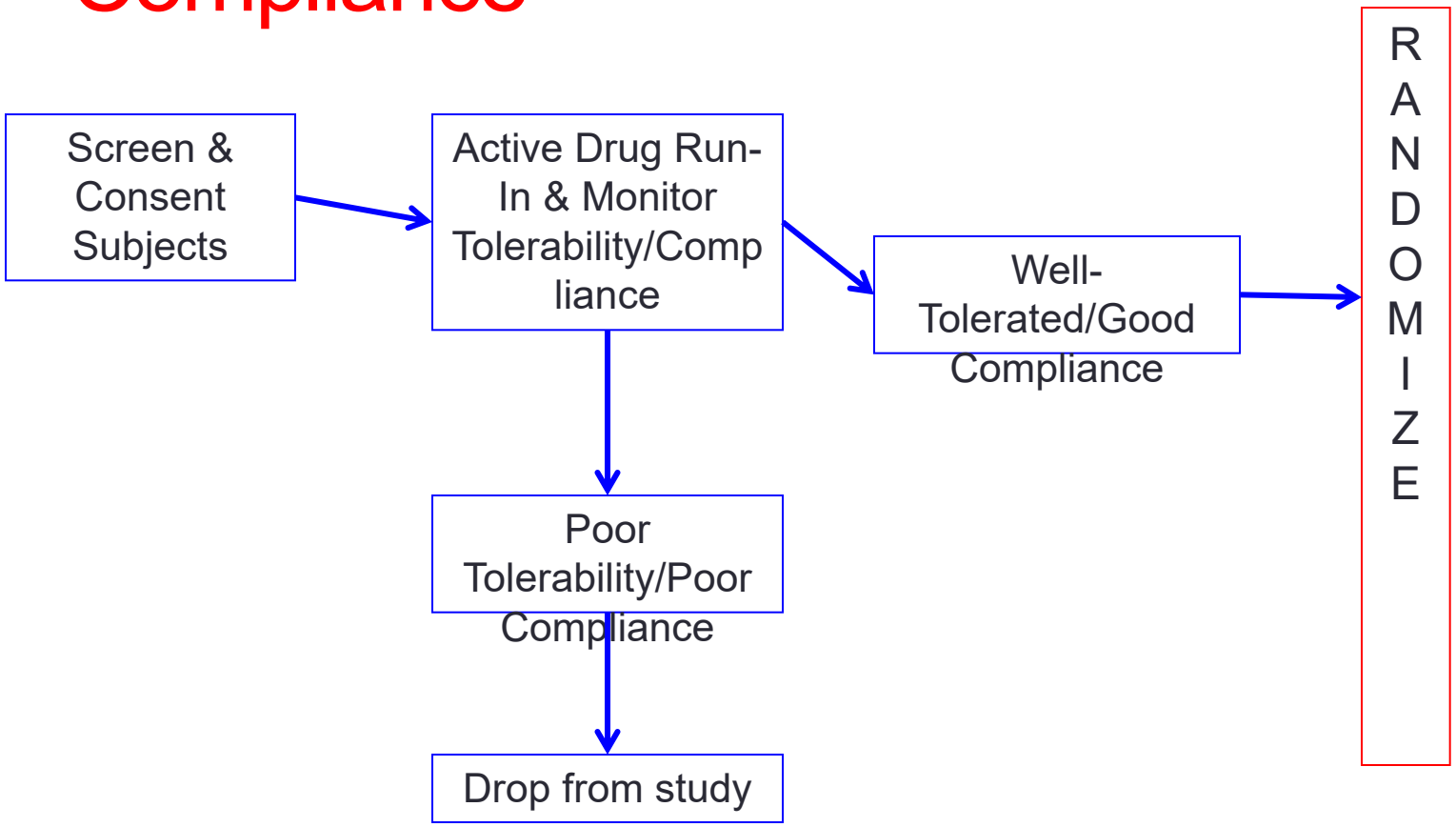
Placebo Run-In to Minimize Non-Compliance



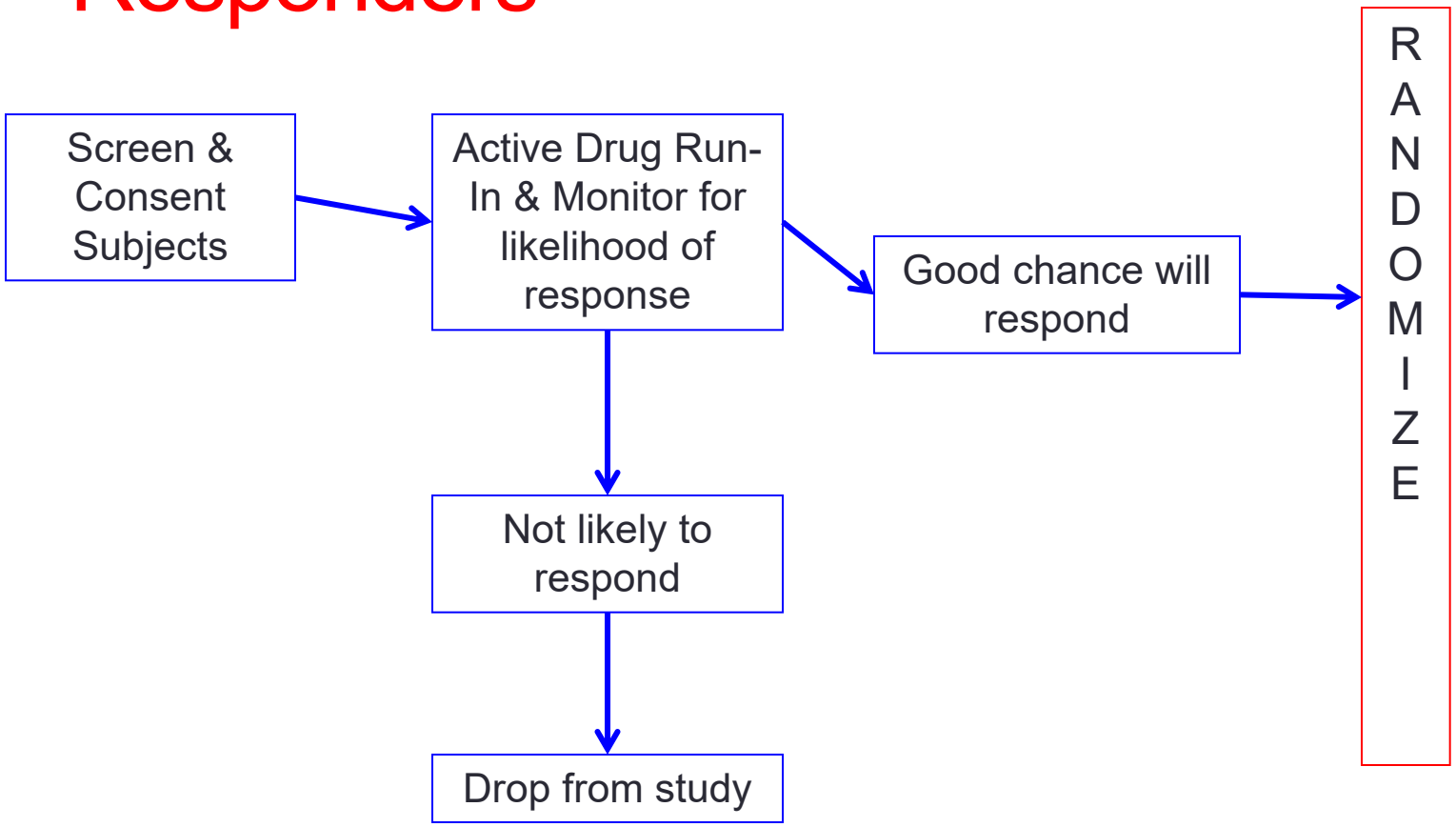
Placebo Run-In to Minimize Placebo Effect



Active Drug Run-In to Minimize Non-Compliance



Active Drug Run-In to Select Likely Responders



RCT Design: Superiority vs. Non-Inferiority Trials

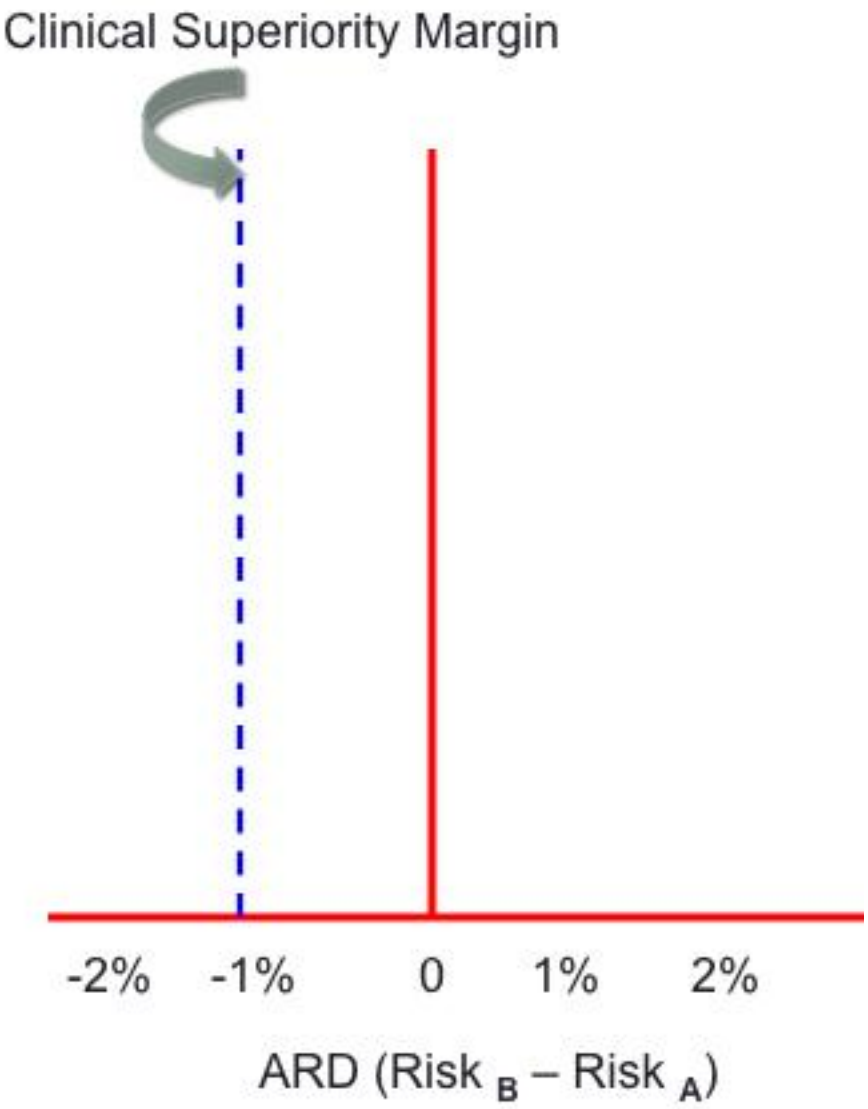
- Superiority Trials:
  - A-Statistical Superiority
  - B-Clinical Superiority

Let's imagine the following situation, two drugs **A** and **B** that are beneficial for prevention of Myocardial Infarction (M.I.). Drug A is the current standard drug used to prevent M.I. and Drug B is under development. Clinical investigators believe that Drug B is more efficacious than drug A and would like to demonstrate its superiority vs. Drug A. The main outcome of the RCT was the risk of M.I. at 30 days following treatment initiation. To be considered as "clinically superior" the Absolute Risk Difference (ARD) of M.I. between B and A should be < -1%.

Superiority clinical trials

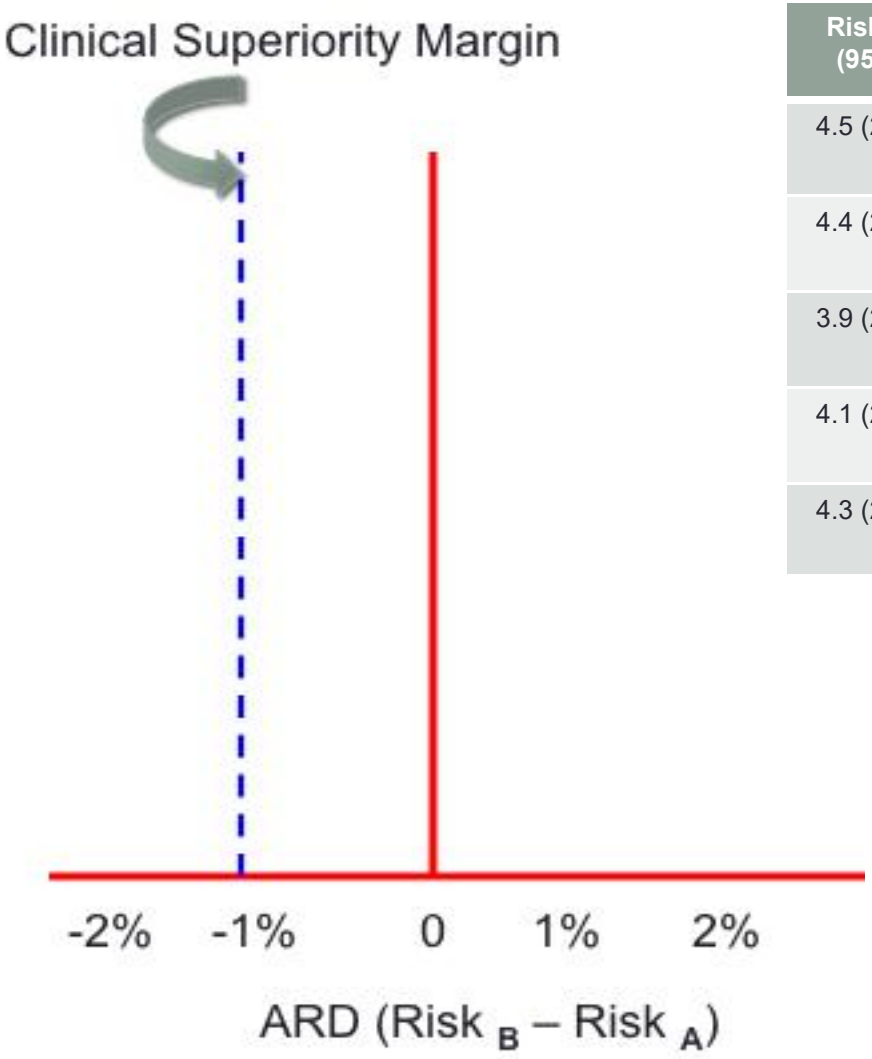
- **Superiority**, equivalence, and non-inferiority **trials**. ... When the aim of the randomized controlled **trial** (RCT) is to show that one treatment is superior to another, a statistical test is employed and the **trial** (test) is called a **superiority trial**(test).
- a **non-inferiority trial** aims to demonstrate that the test product is not worse than the comparator by more than a small pre-specified amount. This amount is known as the **non-inferiority margin**,

RCT Design: Superiority Trial



	H_o	H_a
Statistical Superiority		
Clinical Superiority		

RCT Design: Superiority Trial



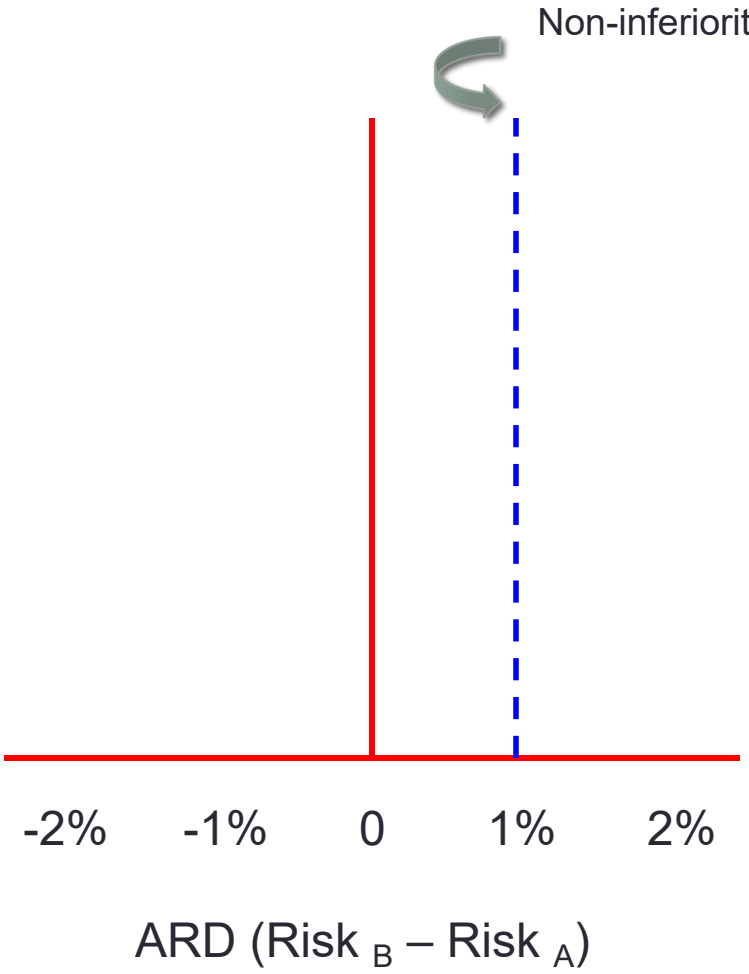
Risk $_B$ (%) (95% CI)	Risk $_A$ (%) (95% CI)	ARD (%) (95%CI)	Clinically Superior?	Statistically Superior?
4.5 (2.5-6.2)	5.9 (4.6-7.3)	-1.4 (-2.1 to -0.9)		
4.4 (2.4-5.9)	5.8 (4.7-7.0)	-1.4 (-2.3 to -1.1)		
3.9 (2.3-4.7)	4.7 (3.5-6.2)	-0.8 (-1.2 to -1.5)		
4.1 (2.6-6.2)	5.1 (4.1-6.0)	-1.0 (-1.5 to 0.2)		
4.3 (2.4-5.1)	5.7 (3.6-6.8)	-1.4 (-1.2 to -1.7)		

RCT Design: Non-Inferiority

- Non-inferiority: The new treatment is "not much worse" compared to control
  - New treatment is not expected to be better on primary efficacy endpoint but is easier to administer/better compliance/more tolerability
  - Has to define a non-inferiority margin

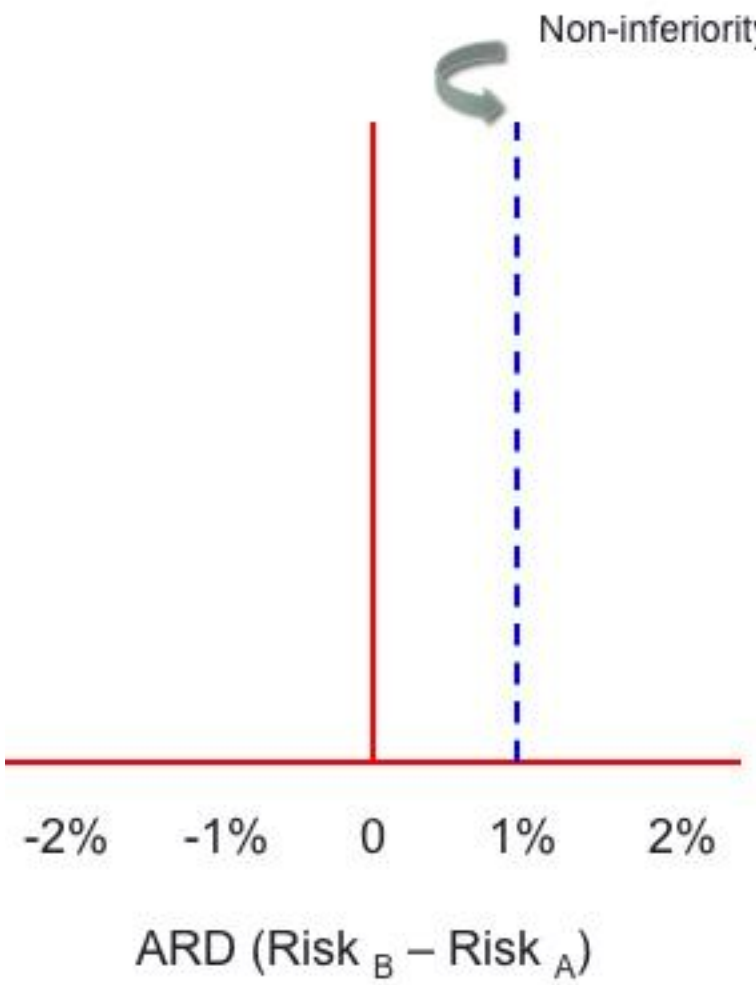
Two drugs **A** and **B** that are beneficial for prevention of Myocardial Infarction (M.I.). Drug A is the current standard drug used to prevent M.I. and Drug B is under development. Clinical investigators believe that Drug B may be not equal drug A in terms of reducing the risk of M.I. but it is easier to administer and may be associated with less bleeding risk. They have elected to go with a non-inferiority design and determined the non-inferiority margin to be that the Absolute Risk Difference (ARD) of M.I. between B and A should not be 1% or higher.

RCT Design: Non-Inferiority



	H_o	H_a
Non-Inferiority Margin (ARD ≥ 1%)		

RCT Design: Non-Inferiority



Risk $_B$ (%) (95% CI)	Risk $_A$ (%) (95% CI)	ARD (%) (95%CI)	Non-Inferior?
6.3 (4.2-8.1)	5.9 (4.6-7.3)	0.4 (-0.4 to 0.8)	
4.4 (2.4-5.9)	5.8 (4.7-7.0)	-1.4 (-2.3 to -1.1)	
6.4 (4.3-8.6)	5.9 (4.6-7.3)	0.5 (-0.3 to 1.3)	
6.7 (5.1-8.2)	5.3 (4.2-6.5)	1.4 (0.9 to 1.7)	

RCT Study Design: Outcomes

- Clinical Outcomes/Endpoints
  - Efficacy and Safety
  - Primary Efficacy and Safety Outcomes
  - Secondary Efficacy and Safety Outcomes

- Surrogate (Intermediate) Outcomes
  - Surrogate endpoints precede clinical endpoints
  - Can predict with confidence the changes in the clinical endpoints
  - e.g. Blood Pressure Control in Anti-Hypertensive Clinical Trials is a surrogate endpoint to Risk of M.I., Risk of HF (Clinical Outcomes)